

2 Multiple Linear Regression

2.1 Introduction

Single Continuous Response $\rightarrow y$

Multiple Predictors $\rightarrow x_1, \dots, x_p$

Each observation $(y_i, x_{i1}, \dots, x_{ip})$, $i = 1, \dots, n$

$i \rightarrow$ day

$y_i \rightarrow$ Total Rainfall on day i .

$x_{i1} \rightarrow$ Total Temp at 6 am on day i .

$x_{i2} \rightarrow$ Relative Humidity at 8AM on day i .

$\vdots$

$x_{ip} \rightarrow$ other factors.

2.2 Linear Model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i, \quad i = 1, \dots, n$$

Assumptions

1. Linear model holds
2. ϵ_i are independent of each other.
3. ϵ_i are independent of $x_{i1}, \dots, x_{ip}$.
4. $E[\epsilon_i] = 0$, $\text{Var}[\epsilon_i] = \sigma^2$. (ϵ_i are continuous r.v.'s)

Matrix Notations

$$\underline{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}_{n \times 1} \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}_{(p+1) \times 1} \quad \underline{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}_{n \times 1}$$
$$X = \begin{pmatrix} 1 & x_{11} & \dots & x_{1p} \\ 1 & x_{21} & \dots & x_{2p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{pmatrix}_{n \times (p+1)}$$

The model can be written as

$$Y = X\beta + \epsilon \quad \rightarrow \quad \epsilon = Y - X\beta$$

Least Squares Criterion

Minimize $\sum_{i=1}^n \epsilon_i^2$ (minimize sum of squares of errors)

$$\begin{aligned} \sum \epsilon_i^2 &= \underline{\epsilon}^T \underline{\epsilon} = (Y - X\beta)^T (Y - X\beta) \\ \text{where } \underline{\epsilon} &= Y - X\beta \\ &= Y^T Y - \beta^T X^T Y - Y^T X \beta + \beta^T (X^T X) \beta \\ S(\beta) &= Y^T Y - 2Y^T X \beta + \beta^T (X^T X) \beta \quad \textcircled{1} \end{aligned}$$

Need to take the derivative of ① w.r.t β ,

$$\frac{\partial S(\beta)}{\partial \beta} = \begin{pmatrix} \frac{\partial S}{\partial \beta_0} \\ \frac{\partial S}{\partial \beta_1} \\ \vdots \\ \frac{\partial S}{\partial \beta_p} \end{pmatrix} = 0 \quad (\text{Normal eq's})$$

Let A be a $1 \times (p+1)$ matrix, $A = (a_0, a_1, \dots, a_p)$.

$$A\beta = \sum_{j=0}^p a_j \beta_j$$

$$\frac{\partial}{\partial \beta}(A\beta) = A^T$$

In our case, $A = \begin{matrix} -2Y^T & X \\ 1 \times n & n \times (p+1) \end{matrix}$, so

$$\frac{\partial}{\partial \beta}(-2Y^T X \beta) = (-2Y^T X)^T = -2X^T Y$$

Now consider the quadratic term with a matrix $B_{(p+1) \times (p+1)}$.

$$B = \begin{pmatrix} b_{00} & b_{01} & \dots \\ b_{10} & & \\ \vdots & & \\ b_{p0} & \dots & b_{pp} \end{pmatrix} = ((b_{ij})) \quad \begin{pmatrix} b_0 \\ b_1 \\ \vdots \\ b_p \end{pmatrix}$$

B is symmetric.

$$\frac{\partial(\beta^T B \beta)}{\partial \beta_i} = \sum_j b_{ij} B_j + \sum_i B_i b_{ij} = 2 \sum_j b_{0j} B_j$$

$$\frac{\partial(\beta^T B \beta)}{\partial \beta_1} = 2b^T \beta$$

$$\begin{bmatrix} \frac{\partial(\beta^T B \beta)}{\partial \beta_0} \\ \frac{\partial(\beta^T B \beta)}{\partial \beta_1} \\ \vdots \\ \frac{\partial(\beta^T B \beta)}{\partial \beta_p} \end{bmatrix} = \begin{bmatrix} 2b_0^T \beta \\ 2b_1^T \beta \\ \vdots \\ 2b_p^T \beta \end{bmatrix} = 2B\beta$$

If $\beta = X^T X$ is symmetric, then

$$\frac{\partial(\beta^T X^T X \beta)}{\partial \beta} = 2X^T X \beta$$

$$[Y^T Y - 2Y^T X \beta + \beta^T (X^T X) \beta] = 0$$

Normal Equations

Thus,

$$-2X^T Y + 2X^T X \beta = 0 \Rightarrow X^T X \beta = X^T Y$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

This is the least squares estimate of β .

Condition: $n > p + 1$. For example, $p = 1, n = 2$.

In regression, there are more data points than predictor dimensions; otherwise, we can have a perfect fit with all $\hat{\varepsilon}_i = 0$, which will be the least squares solution, $\hat{\beta} = X^T Y$ if $n = p + 1$. All the formulation of least squares or best line have to be done because we have more points than dimensions.

$$Y = X\beta + \varepsilon$$

Residuals are $y_i - \hat{y}_i = \hat{\varepsilon}_i$.

Fitted values are $\hat{y}_i = \beta_0 + \beta_1 x_i + \dots + \beta_p x_{ip}$.

2.3 Diagnostics

Assumptions need to be verified for the errors.

We verify them for the residuals. - Plot residuals against fitted values of each predictor. (Linear model dependence on x_i)

- Plot squared residuals against fitted values of each predictor (Variance is same, check for homoscedasticity). - Normal Q-Q plot if normality is assumed. (outliers, influential observations, leverage)

2.4 Expectation, Variance, Covariance of matrix of $\hat{\beta}$

$$\hat{\beta} = (X^T X)^{-1} X^T Y, \quad Y = X\beta + \varepsilon$$

$$E[X] = \mu, \quad Var(X) = \Sigma$$

$$E[AX + b] = AE[X] + b$$

$$Var(AX + b) = AVar(X)A^T$$

$$E[\hat{\beta}] = \beta + (X^T X)^{-1} X^T E[\varepsilon] = \beta$$

$$Var(\hat{\beta}) = (X^T X)^{-1} X^T Var(\varepsilon) X (X^T X)^{-1}$$

If $Var(\varepsilon) = \sigma^2 I$, then

$$Var(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$$

Variance Properties

If $Ax = \sum_{j=1}^K a_{ij} x_j$, then

$$Var(Ax + By) = a^2 Var(x) + b^2 Var(y) + 2ab Cov(x, y)$$

For a random vector X and a matrix A :

$$Var(AX) = AVar(X)A^T$$

2.5 Sum of Squares Decomposition

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$SSR = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n \varepsilon_i^2$$

$$= Y^T Y - 2Y^T X \hat{\beta} + \hat{\beta}^T (X^T X) \hat{\beta}$$

Substituting $\hat{\beta} = (X^T X)^{-1} X^T Y$:

$$SSR = Y^T (I - X(X^T X)^{-1} X^T) Y$$

Degrees of freedom:

Source	SS	df
Regression	SS_{Reg}	p
Residual	SS_{Res}	$n - (p + 1)$
Total	SST	$n - 1$

$$\hat{\sigma}^2 = \frac{SSR}{n - (p + 1)}$$

$$R^2 = 1 - \frac{SSR}{SST}, \quad \text{Adjusted } R^2 = 1 - \frac{SSR/(n - (p + 1))}{SST/(n - 1)}$$

As more predictors are added, SSR decreases.

2.6 Other issues

1. $Y = \beta_0 + \beta_1 \log x + \beta_2 x^2 + \epsilon \rightarrow$ A linear model

$$x = \frac{\beta_1 X}{\beta_0 + X} + \epsilon \quad \text{is a non linear.}$$

The 2nd one can't be expressed as a linear combination of known functions of X.

2. We are only dealing with models, which is not necessarily the truth.

“All models are wrong but some models are useful” – Box (1976)

Not even looking for correct or true.

3. We do not claim a causal relation between X & Y.

i.e. Trying to predict Y based on X.

Not claiming X causes Y or Y causes X.

4. Interpretation of the coefficients and Simpson's Paradox

Univariate ($p = 1$) regression $Y = \beta_0 + \beta_1 X + \epsilon$

- $\beta_0 \rightarrow$ expected value of Y when $X = 0$.
- $\beta_1 \rightarrow$ avg change in Y for unit change in X.

Horse Power (X_2) in car weights (X) For every value of X_2 fixed, Y is also a function of X_1 .

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon, \quad \beta_1 > 0.$$

If we disregard X_2 , then the overall pattern of Y on X_1 is increasing.

In decreasing $Y = \beta_0 + \beta_1 X_1 + \epsilon$, $\beta_1 < 0$.

In multiple regression, the interpretation of β_1 is the avg change in Y for unit change in X_1 , when X_2 is held constant.

In general the regression coefficient measures the change in response for unit change in the corresponding predictor when all other predictors are held constant.

5. Collinearity A linear dependence b/w 2 or more predictors/columns of the design matrix.

$$X = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ 1 & x_{21} & \dots & x_{2p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix}_{n \times p}$$

e.g. $x_1 = 2x - 7$

$$x_1 = x_2 + 2x_3$$

At the population level makes the coeff $(\beta_1, \beta_2, \beta_3)$ ill defined.

$$y = x_1 + x_2 + \epsilon \quad (1, 1, 1)$$

$$y = 3 - 2x_2 + \epsilon \quad (0, 3)$$

$$y = 1.5x_1 + \epsilon \quad (1.5, 0)$$

Infinitely many β 's represent same plane.

At the estimate level The OLS estimate is $\hat{\beta} = (X^T X)^{-1} X^T Y$. If $x_1 = 2x_2$ then $(X^T X)$ is not full rank. As $\det(X^T X) = 0 \rightarrow$ then $(X^T X)^{-1}$ can't be defined.

Let $\det(X^T X)$ not be zero, but very small, then $(X^T X)^{-1}$ becomes unstable conceptually. This is called near collinearity & we should try to avoid. We want to remove some of the columns. But which ones?

- If the relation involves only 2 elements, then a pairwise plot e.g., (x_1, x_2) , (x_1, x_3) , (x_2, x_3) , will reflect an exact straight line in one of the plots. Then drop any one of the variables involved in that plot.
- Even a very high correlation b/w (x_1, x_3) dictates that we drop one of them.
- If $x_1 = x_2 + 2x_3 \rightarrow$ This can't be detected from a particular plot. One must check the determinant of sub-matrices. For predictors e.g., (X_1, X_2, X_3, X_4) , analyze $(X^T X)^{-1}$.

Each diagonal element of the inverse is related to the determinant of the minor of the original matrix after removing that particular row and column.

- The columns **not** involved in the linear relation will have small numbers in the corresponding diagonal of the inverse matrix.
 - Retain those.
- From the subset that is not retained, remove elements one by one & check if the overall determinant stabilizes.
- **Collinearity is:** Linear or approximate linear relationship between the predictor variables.
- **Why is it a problem?** The estimate $\hat{\beta}$ is unstable, i.e., the variance $(X^T X)^{-1} \sigma^2$ is very large.
- **How to detect?** $\det(X^T X) \approx 0$. Also can be done by using a pairs plot.

6. Interaction The nature of a relationship between y & X_1 depends on the value of X_2 .

Our assumption is that every variable makes a distinct additive contribution to the response. The interaction model is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon$$

Start with a bigger (with interaction) model. Test the hypothesis whether the interaction term is zero. If the hypothesis ($H_0 : \beta_3 = 0$) is not rejected, then we can go ahead with the simpler model (with no interaction).

7. Normal Equations & the Geometry of Least-Squares The model is $Y = X\beta + \epsilon$. We find the coefficients by minimizing $\sum_{i=1}^n \epsilon_i^2 = \|\epsilon\|^2$.

Geometrically, $Y \in R^n$, and the fitted values $X\beta$ form a plane (or subspace) in that n -dimensional space. The distance is minimized when we drop a perpendicular/normal from the point Y to the plane.

The normal equations are derived from this condition:

- The residual vector ϵ must be orthogonal to the column space of X : $X^T \epsilon = \mathbf{0}$.
- Substituting $\epsilon = Y - X\beta$, we get: $X^T(Y - X\beta) = \mathbf{0}$.
- This gives the final form: $X^T Y = X^T X \beta$.